

User Guide for Pet Fostering Database System

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1. Applicable Scenarios

This system simulates a professional pet care center that focuses on providing pet boarding and standardized training services to pet owners (clients). The center offers services for common pet types, such as dogs and cats, providing categorized cages (small dog cages, medium-sized dog cages, large dog cages, cat cages), each with a fixed price and quantity limit; it also offers pre-set training programs (such as basic training for puppies, behavior correction for adult dogs, interactive training for cats, etc.). The core of the center's daily operation lies in matching customer needs with service resources: verifying the valid vaccination certificates required for pet boarding, allocating appropriate cages and trainers with the corresponding qualifications, managing the entire lifecycle of service reservations and orders, calculating fees based on service types, and handling reservation cancellations or related fines for not showing up. The core goal of this system is to ensure the efficient and conflict-free implementation of services, simplify the administrative processes of the center, and provide customers with a transparent service tracking channel.

Supported processes:

1. Customer and pet information management (including registration, update and storage of customer contact information, details of pet breeds/birth dates, and vaccination records, etc., to meet the requirements for adoption eligibility);
2. Service reservation and order creation (for each pet, a separate order needs to be created for each service. The system will automatically check for time conflicts to avoid scheduling the same pet for foster care services and training services simultaneously);

3. Resource allocation (selecting appropriate cage types based on the pet's size and breed; only qualified trainers will be assigned training programs);
4. Training course management (setting exclusive service rules for one-on-one trainers and setting the upper limit of 5 pets per training group; tracking the participation numbers and status of group courses);
5. Order lifecycle management (reservations for accommodation and care services need to be completed 24 hours in advance, and cancellations need to be notified 12 hours in advance. If the cancellation is not made on time and the items are not retrieved from the care institution on time, 50% of the fee will be deducted; during accommodation and care, if additional training services are needed, the training time must not overlap with the fixed accommodation and care arrangements);
6. Cost accounting and payment processing (boarding fees are calculated as "number of days × cage unit price", training fees are calculated as "number of courses × project unit price"; record payment methods, amounts and status);
7. Basic operation tracking and query (query availability status of training periods, service records of trainers, customer order history, progress of training courses and payment status, etc.).

2.Main Functions

- Pet owner information management
- Pet file management
- Vaccination record tracking
- Service reservation management
- Cage usage management
- Training progress tracking
- Payment processing

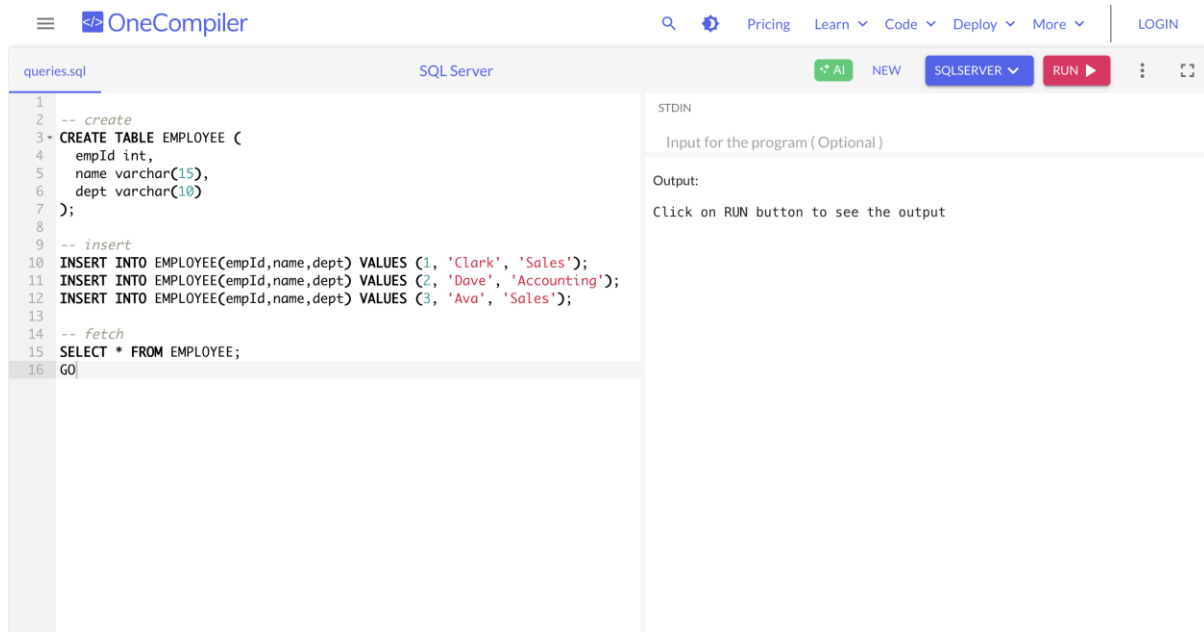
3.Database Set Up

Considering that database users lack a certain degree of professionalism in application, we adopt an online SQL compiler called OneCompiler (<https://onecompiler.com/sqlserver>). This compiler is quite user-friendly as it may provide the results directly on the attached column next to the

code.



Wait for the page to load completely, and you will see the SQL editor and the Run button.



You can set up the database by using the following steps:

- (1) Copy the complete table creation SQL script and paste it into the editor of OneCompiler.
- (2) Click the "Run" button to execute

Important Note: The OneCompiler environment is temporary. Data may be lost after closing the browser. It is recommended to back up the SQL script before performing important operations. Each new session requires re-running the table creation script.

3. Instructions in using the key functionalities of the application

(1) Checking all information about pets

```
SELECT p.Pet_ID, p.Name, p.Species, p.Breed, po.First_Name, po.Last_Name
FROM Pet p
JOIN Pet_Owner po ON p.Owner_ID = po.Owner_ID;
```

(2) Checking availability of the cages

```
SELECT TypeName, AvailableQuantity
FROM Cage
WHERE AvailableQuantity > 0;
```

(3) Inserting new data (e.g. New Pet_Owner)

```
INSERT INTO Pet_Owner (Owner_ID, First_Name, Last_Name, Phone, Email, Address,
Registration_Date)
VALUES (9, 'Xiao', 'Chen', '13800138009', 'chenxiao@example.com', 'Beijing Road,
Guangzhou', '2025-11-20');
To se the result, you can:
SELECT * FROM Pet_Owner WHERE Owner_ID = 9;
```

(4) Deleting data (e.g. Updating pet information)

```
UPDATE Pet SET Weight = 24.0 WHERE Pet_ID = 13 AND Owner_ID = 9;
Check:
SELECT * FROM Pet WHERE Pet_ID = 13;
```

4.Sample Analysis Report

This is a sample of some important and complex queries, which might be crucial for the business of a pet fostering agency; you may make this a reference to the business analysis.

(1) Training effectiveness evaluation

```
SELECT tp.TrainingDate AS 'Training Date', tp.SkillsCovered AS 'Training Content',
tp.TrainingRating AS 'Rating', p.Name AS 'Pet Name', e.FirstName + ' ' + e.LastName AS
'Trainer'
FROM TrainingProgress tp
JOIN Booking b ON tp.BookingID = b.Booking_ID
JOIN Pet p ON b.Pet_ID = p.Pet_ID AND b.Owner_ID = p.Owner_ID
JOIN Employee e ON b.Employee_ID = e.EmployeeID
```

ORDER BY tp.TrainingRating DESC;

Output:

Training Date	Training Content
---------------	------------------

2025-10-08	Sit, Shake Hands, Heel
2025-11-10	Basic Command Introduction, Interactive Cooperation
2025-10-19	Leash Training, Socialization
2025-10-15	Down, Stay, Recall
2025-10-12	No Jumping, No Chewing, Quiet Command

Business Insights:

Top Performing Trainer: Coach Chen achieves the highest ratings (5/5) for basic obedience training.

Most Effective Training Content: Basic commands like "Sit" and "Shake Hands" receive perfect scores.

Improvement Opportunity: Behavior correction training shows lower ratings (3/5), suggesting a need for curriculum enhancement.

Customer Satisfaction: High ratings indicate well-structured training programs and skilled trainers.

(2) Checking the using report of cages

```
SELECT
    TypeName AS 'Cage Type',
    TotalQuantity AS 'Total Quantity',
    AvailableQuantity AS 'Available Quantity',
    (TotalQuantity - AvailableQuantity) AS 'Used Quantity',
    ROUND(((TotalQuantity - AvailableQuantity) * 100.0 / NULLIF(TotalQuantity, 0)), 2) AS
    'Used Rate(%)'
FROM Cage
ORDER BY 'Used Rate(%)' DESC;
```

Output:

Cage Type	Total Quantity Available	Quantity Used	Quantity Used Rate(%)
large dog cage	4	2	50.0000000000
cat cage	5	3	40.0000000000
medium dog cage	8	5	37.5000000000
small dog cage	6	4	33.3300000000

Analysis:

Highest Demand: Large dog cages show 50% utilization rate, indicating strong demand for big dog boarding.

Optimal Capacity Management: Current cage quantities appear well-balanced with no overcrowding.

Expansion Opportunity: Medium dog cages have the highest total usage (3 cages), suggesting potential for adding more units.

Resource Planning: Consistent 30-50% utilization across all cage types shows healthy business volume.

(3) Tracking the most popular services

```
SELECT
    s.Service_Name AS 'Service Name',
    COUNT(b.Booking_ID) AS 'Booking Times',
    SUM(p.Amount) AS 'Total revenue',
    AVG(CAST(s.Service_Rating AS DECIMAL(10,2))) AS 'Average Rate'
FROM Service s
LEFT JOIN Booking b ON s.Service_ID = b.Service_ID
LEFT JOIN Payment p ON b.Booking_ID = p.BookingID
GROUP BY s.Service_ID, s.Service_Name
ORDER BY COUNT(b.Booking_ID) DESC;
```

Output:			
Service Name	Booking Times	Total revenue	Average Rate
Pet Boarding (Medium Dog Cage)	5	1350.00	5.000000
Pet Boarding (Cat Cage)	4	720.00	5.000000
Puppy Basic Training	2	1800.00	5.000000
Pet Boarding (Small Dog Cage)	2	420.00	4.000000
Adult Dog Behavior Correction	1	1200.00	4.000000
Pet Boarding (Large Dog Cage)	1	360.00	4.000000

Business Insights:

Most Popular Service: Medium dog boarding leads with 5 bookings and ¥1,350 revenue

Highest Revenue Generator: Training services generate premium income (¥1,800 for puppy training).

Customer Satisfaction: All services maintain 4+ star ratings, with boarding services achieving perfect 5-star ratings.

Service Mix Strategy: Boarding services drive volume while training services provide higher-margin revenue.

Market Demand: Strong preference for medium-sized dog accommodations suggests customer demographic trends.